Binary Tree Representation & Types Data Structures

> **Definition:** A **Binary Tree** is a hierarchical tree data structure in which every node has at most **two children**, referred to as the **left child** and the **right child**.

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1. Detailed Technical Explanation

1. Types of Binary Trees

```

1. FUL

3. PERFECT BINARY TREE 4. DEGENERATE / SKEWED TREE
[ 1 ]
```

1. **Full (Proper/Strict) Binary Tree:** Every node has either 0 or 2 children. No node has only 1 child.

2. **C
gaps.

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2. Binary Tree Representations in Memory

1. Sequential (Array-Based) Representation

For a complete binary tree stored in array `A` (1-indexed):

- No

2. Linked (Pointer-Based) Representation in C

```c
#incluc

// Structure for a Binary Tree Node
struct T

// Function to create a new tree node
struct T
```

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3. Mathematical Properties of Binary Trees

- Maximum nodes at level `i` (root at level 0) = `2^i`.
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4. Must-Write Points for Exams

- In a Full Binary Tree, no node has degree 1.
- In a

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5. Quick Recall Flow

```
Binary
```